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# Semantic Region-Guided Transferable Attacks on Vision-Language Pretraining Models

Jiayang Pan, Chen Wan, Wutao Chen, Lifeng Huang

***Abstract*—Vision-language pre-trained (VLP) models have achieved strong performance on multimodal tasks, but they remain vulnerable to adversarial attacks. In black-box settings, transferability is often limited because perturbation generation relies on surrogate-specific saliency cues that generalize poorly across VLP architectures. In this letter, we propose Semantic Region-Guided Attack (SRGA), a transferable attack framework that replaces such cues with detector-derived semantic regions to provide more consistent cross-model guidance. Based on these regions, SRGA performs coordinated perturbation generation in both image and text modalities with lightweight regional transformations and spatially weighted optimization. Experiments on Flickr30K and MSCOCO show that SRGA achieves competitive black-box transferability across diverse VLP architectures under multiple settings, with modest additional cost.**

***Index Terms*—Adversarial example, black-box attack, transferability, semantic region guidance.**

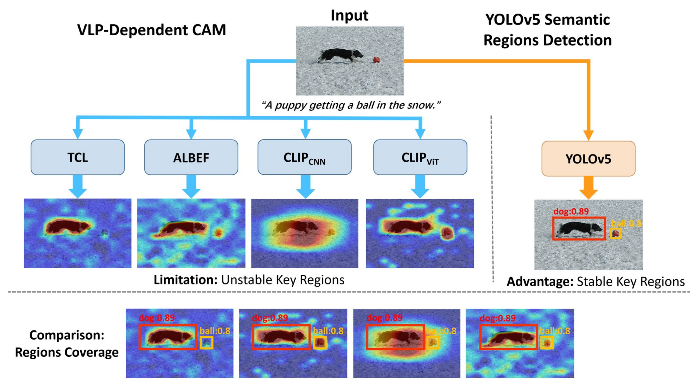


Fig. 1: Comparison of Grad-CAM maps from different VLP models and semantic regions produced by an object detector.

## I. Introduction

VISION-language pre-trained (VLP) models have achieved strong performance across a wide range of multimodal tasks [1], [2], but they remain vulnerable to adversarial examples [3], [4]. Even small perturbations to either modality can disrupt cross-modal semantic alignment [5], [6]. Among the properties of adversarial examples, transferability across models is especially important in black-box settings, where the target model is inaccessible [7], [8], [9], [10]. This makes transferable attacks particularly relevant for robustness evaluation of multimodal systems [11], [12], [13].

Several methods have been proposed to improve adversarial transferability [14], [15], [16], [17]. Early attacks focused on a single modality, such as PGD [18] for images and BERT-Attack [19] for text. Co-Attack [3] moved beyond single-modality attacks by jointly optimizing image and text perturbations to disrupt cross-modal alignment. More recent methods have further improved transferability through input diversity and perturbation evolution [20], [21], [22], [23]. For example, SGA [4] exploits multi-scale image-text variants, DRA [24] introduces diversity along adversarial trajectories, and SA-AET [5] incorporates semantic alignment into perturbation evolution. However, these methods often rely on surrogate model-specific cues, such as gradients, attention maps, or saliency signals, to guide perturbation generation. These cues can vary substantially across VLP architectures, making the perturbation focus learned on one model less reliable for transfer to others [2], [12], [25], [26]. As shown in Fig. 1, Grad-CAM [27] produces noticeably different saliency maps for the same input across VLP models, suggesting that surrogate-specific cues may not provide reliable guidance for cross-model attacks. This limitation is further reflected in Table I, where detector-derived regions consistently yield stronger transferability than random regions and generally outperform Grad-CAM-guided regions. The comparison with random and Grad-CAM-guided regions suggests that the gain does not come merely from using a local prior, but from adopting detector-derived semantic regions as a more consistent basis for cross-model guidance. These results motivate detector-derived semantic regions as a more reliable basis for transferable attack generation.

In this letter, we propose Semantic Region-Guided Attack (SRGA), a transferable adversarial framework that replaces surrogate-specific saliency cues with detector-derived semantic regions for cross-model guidance. As shown in Fig. 2, SRGA first identifies object-centric regions using an off-the-shelf detector and associates them with corresponding textual entities. It then performs coordinated perturbation generation in both image and text modalities, with regional transformations and spatial weighting used to strengthen perturbations around the detector-grounded region. This design provides more consistent perturbation guidance across VLP architectures and improves black-box transferability. The contributions

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The source code is provided in https://github.com/BeerDuck-envs/SRGA.

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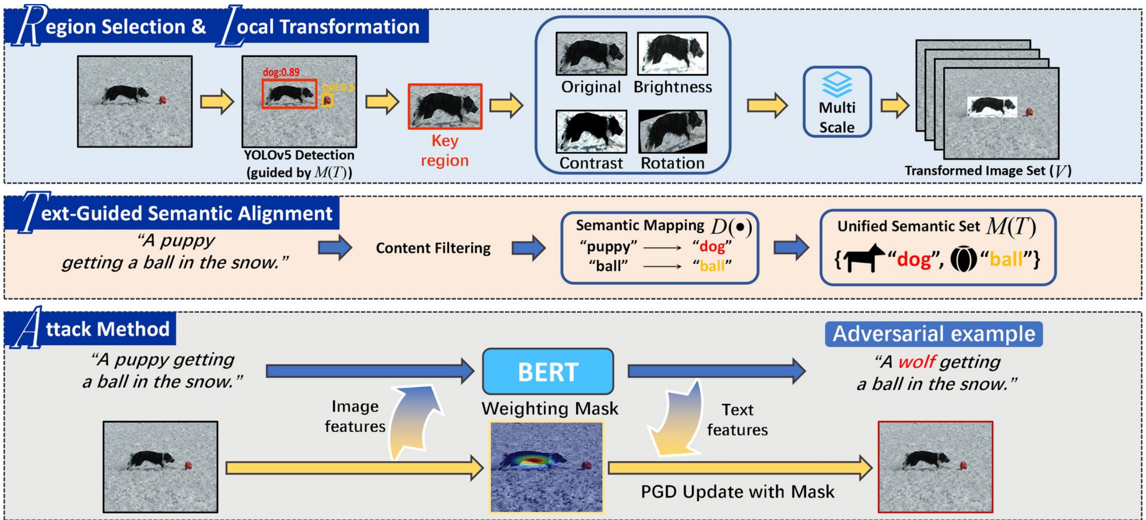


Fig. 2: Overview of the proposed SRGA framework. Detector-derived object regions are aligned with text tokens to guide cross-modal perturbations for disrupting alignment and improving transferability.

TABLE I: Attack success rates (%) under different region guidance strategies.

| Region | ALBEF TR R@1 | ALBEF IR R@1 | TCL TR R@1 | TCL IR R@1 | $\text{CLIP}_{\text{ViT}}$ TR R@1 | $\text{CLIP}_{\text{ViT}}$ IR R@1 | $\text{CLIP}_{\text{CNN}}$ TR R@1 | $\text{CLIP}_{\text{CNN}}$ IR R@1 |
|---|---|---|---|---|---|---|---|---|
| Random | 44.62* | 56.12* | 21.07 | 35.70 | 29.87 | 40.84 | 36.29 | 45.98 |
| Grad-CAM | **60.08*** | **70.23*** | 24.19 | 37.62 | 31.46 | 44.07 | 37.68 | 47.15 |
| YOLOv5 | 59.12* | 68.19* | **26.03** | **39.50** | **33.99** | **46.04** | **38.19** | **48.92** |
| Random | 18.80 | 34.12 | 46.68* | 57.79* | 31.10 | 42.94 | 34.26 | 44.29 |
| Grad-CAM | 23.36 | 38.59 | 62.49* | 69.79* | 32.11 | 45.26 | 37.61 | 47.47 |
| YOLOv5 | **25.34** | **40.18** | **62.70*** | **69.98*** | **33.99** | **46.04** | **38.83** | **48.34** |
| Random | 6.94 | 23.45 | 11.02 | 25.41 | 69.12* | 70.95* | 39.65 | 41.15 |
| Grad-CAM | 10.95 | 27.76 | 13.17 | 28.45 | 74.05* | 77.82* | 41.51 | 46.55 |
| YOLOv5 | **11.68** | **27.92** | **14.01** | **29.40** | **76.44*** | **79.67*** | **42.53** | **48.40** |
| Random | 6.59 | 21.58 | 11.26 | 27.05 | 35.56 | 41.45 | 79.51* | 77.48* |
| Grad-CAM | 8.01 | 26.01 | 13.39 | 28.26 | 37.06 | 45.30 | **84.51*** | **83.40*** |
| YOLOv5 | **9.38** | **26.54** | **14.23** | **29.07** | **38.16** | **45.72** | 84.04* | 83.12* |

are summarized as follows.

1) To the best of our knowledge, this is the first work to use detector-derived semantic regions as an explicit guidance cue for transferable attacks on VLP models.
2) We propose SRGA, a transferable adversarial framework that uses detector-derived semantic regions to guide coordinated perturbations in both image and text modalities.
3) Experiments on multiple VLP architectures demonstrate that SRGA improves black-box transferability with favorable efficiency.

## II. Methodology

### A. Notations

Let $(v,t)$ denote an image–caption pair, where $v \in R^{H\times W\times 3}$ is the input image and $t = \{w_1, \dots, w_n\}$ is the caption token sequence. Let $\mathcal{F}_I(\cdot)$ and $\mathcal{F}_T(\cdot)$ denote the image and text encoders, respectively. Their $\ell_2$-normalized embeddings are defined as $\hat{f}_I(v) = \frac{\mathcal{F}_I(v)}{\|\mathcal{F}_I(v)\|}$ and $\hat{f}_T(t) = \frac{\mathcal{F}_T(t)}{\|\mathcal{F}_T(t)\|}$. Cross-modal similarity is measured by the inner product $\langle \hat{f}_I(v), \hat{f}_T(t)\rangle$. The feasible perturbation sets are defined as $\mathcal{B}[v,\varepsilon_v] = \{v' : \|v'-v\|_\infty \le \varepsilon_v\}$ and $\mathcal{B}[t,\varepsilon_t] = \{t' : d_{\text{tok}}(t',t) \le \varepsilon_t\}$, where $d_{\text{tok}}(\cdot,\cdot)$ counts token substitutions. The goal is to generate $(v',t')$ within these sets to minimize the cross-modal similarity.

### B. Detector-Guided Semantic Region Modeling

Given an input image $v$, a pretrained object detector produces a set of candidate regions

$$\mathcal{D} = \{(b_i, c_i, s_i)\}_{i=1}^{N_b}, \tag{1}$$

where $b_i = (x_i^a, y_i^a, x_i^b, y_i^b)$ denotes the $i$-th bounding box, $c_i \in \mathcal{C}_{\text{det}}$ the predicted category, and $s_i \in [0,1]$ the confidence score. Our framework is not tied to a specific detector architecture. In this work, YOLOv5 serves as a fixed external pretrained detector. It is independent of the target VLP model, is not retrained or adapted for it, and its detections are computed once on the clean image and kept fixed throughout the attack iterations.

To retain text-relevant detections, we define

$$\mathcal{D}_{\text{f}} = \{(b_i, c_i, s_i) \in \mathcal{D} \mid c_i \in \mathcal{M},\ s_i \ge \zeta\}, \tag{2}$$

where $\mathcal{M}$ denotes the text-derived concept set defined in Sec. II-C. The semantic target region is selected as

$$b^* = \begin{cases} \arg\max_{(b_i,c_i,s_i)\in\mathcal{D}_{\text{f}}} \text{Area}(b_i), & \mathcal{D}_{\text{f}} \neq \emptyset, \\ (0,0,W,H), & \text{otherwise}, \end{cases} \tag{3}$$

where $\text{Area}(b_i) = (x_i^b - x_i^a)(y_i^b - y_i^a)$, and $(0,0,W,H)$ denotes the full-image region. This rule provides a simple way to retain a text-relevant object while avoiding small or unstable detections. When no matched region is available, the full image is used as a fallback so that the attack remains well-defined without relying on an unrelated detector region.

Let $p^* = \text{Crop}(v, b^*)$ denote the selected patch. To increase local diversity while preserving the global scene, we apply a local transformation (LT) defined as

$$A(p^*) = \begin{cases} p^*, & \tau = 0, \\ \text{clamp}(p^* + a, 0, 1), & \tau = 1, \\ \text{clamp}((p^* - \bar{p}^*)\beta + \bar{p}^*, 0, 1), & \tau = 2, \\ R(p^*, \theta), & \tau = 3, \end{cases} \tag{4}$$

where $\tau \sim \text{Uniform}\{0,1,2,3\}$ selects identity, brightness, contrast, or rotation, $a \sim \mathcal{U}(-\delta,\delta)$, $\beta \sim \mathcal{U}(0.8,1.2)$, $\theta \sim \mathcal{U}(-5^\circ,5^\circ)$, and $\bar{p}^*$ denotes the mean patch intensity. We further construct intensity-scaled variants

$$v_i^{(s)} = T_{\lambda_i}(v_s'), \qquad \mathcal{V}_s = \{v_i^{(s)}\}_{i=1}^{K}, \tag{5}$$

where $T_{\lambda_i}(\cdot)$ applies $A(\cdot)$ within $b^*$ and rescales the transformed patch by $\lambda_i$. These variants increase local diversity around the detector-grounded region while preserving the global image structure.

### C. Text-Guided Region Matching

To associate text with detector-derived regions, we extract noun tokens from the input text $t = \{w_1,\ldots,w_n\}$ using part-of-speech tagging:

$$\mathcal{N} = \{w_i \mid w_i \in t,\ w_i \text{ is a noun}\}. \tag{6}$$

We then introduce a Semantic Normalization Dictionary (SND), denoted by $D(\cdot)$, to reduce lexical mismatch between caption nouns and detector labels. Built on the 80 categories of the pretrained YOLOv5 detector, SND is constructed by aligning caption nouns from Flickr30K and MSCOCO with detector categories through LLM-assisted semantic matching. The resulting dictionary contains 3382 noun-to-category mappings and is fixed across experiments. The alignment set is

$$\mathcal{M} = \{D(w_i) \mid w_i \in \mathcal{N},\ D(w_i) \in \mathcal{C}_{\text{det}}\}, \tag{7}$$

where $D(w_i)$ denotes the normalized concept of token $w_i$. The resulting set $\mathcal{M}$ provides a lightweight semantic bridge between caption entities and detector outputs for selecting text-relevant visual regions.

### D. Attack Method

SRGA first perturbs the text to create an initial semantic mismatch, and then optimizes the image under the perturbed caption, with updates biased toward the detector-grounded region. This separates discrete textual perturbation from continuous visual optimization.

*1) Textual Perturbation:* Given $(v,t)$, the adversarial caption is generated by maximizing the negative image-text similarity, which is equivalent to minimizing the cross-modal similarity.

$$t' = \arg\max_{t' \in \mathcal{B}[t,\varepsilon_t]} \left(-\langle \hat{f}_T(t'), \hat{f}_I(v)\rangle\right). \tag{8}$$

In practice, we adopt BERT-Attack to generate candidate substitutions and greedily select the words that maximize Eq. (8) under the token budget $\varepsilon_t$.

*2) Image Perturbation:* With $t'$ fixed, the image-side objective is defined over the transformed variants in $\mathcal{V}_r$:

$$\mathcal{L} = -\frac{1}{K}\sum_{i=1}^{K}\langle \hat{f}_T(t'), \hat{f}_I(v_i^{(r)})\rangle. \tag{9}$$

To emphasize the semantic region without suppressing perturbations elsewhere, we define a Weighting Mask (WM) as

$$W(x,y) = 1 + \exp\left(-\frac{(x-x_c)^2 + (y-y_c)^2}{2\sigma^2}\right), \tag{10}$$

where $W(x,y)$ is broadcast over image channels, $(x_c,y_c)$ is the center of $b^*$, and $\sigma = \rho\min(w_{b^*},h_{b^*})$ controls spatial decay. Since $W(x,y) = 1+\exp(\cdot)$, it takes values approximately in $(1,2]$, assigning larger weights near the center of $b^*$ while retaining nonzero support over the full image.

Starting from $v_0' = v$, projected gradient descent updates are performed as

$$v_{r+1}' = \Pi_{\mathcal{B}[v,\varepsilon_v]}\left(v_r' + \eta\, W(x,y) \odot \text{sign}(\nabla_{v_r'}\mathcal{L})\right), \tag{11}$$

where $\eta$ is the step size, $\odot$ denotes element-wise multiplication, and $\Pi_{\mathcal{B}[v,\varepsilon_v]}(\cdot)$ projects the updated image back to the $\ell_\infty$ ball. During image optimization, the PGD update is computed from the average loss of transformed variants, with the weighting mask emphasizing the detector-grounded region.

## III. Experiments

### A. Experimental Settings

Our method is implemented in PyTorch and primarily evaluated on an NVIDIA RTX 4090D GPU.

**Datasets**. The proposed method is evaluated on two multimodal benchmarks, Flickr30K [28] and MSCOCO [29]. Flickr30K contains 31,783 images, each annotated with five captions, while MSCOCO includes 123,287 images with five captions per image on average.

**Models**. We consider two representative categories of VLP models: fused models with joint embeddings (ALBEF [30] and TCL [31]) and aligned models with separate encoders under contrastive supervision ($\text{CLIP}_{\text{ViT}}$ and $\text{CLIP}_{\text{CNN}}$ [32]).

**Baselines**. To verify the effectiveness of the proposed method, we compare it with various approaches, including PGD [18], BERT-Attack [19], Sep-Attack, Co-Attack [3], SGA [4], DRA [24], and SA-AET [5].

**Attack Settings**. For textual attacks, we employ BERT-Attack [19] with a perturbation budget of $\varepsilon_t = 1$ and candidate size $k = 10$. For image attacks, we use PGD [18] with a perturbation budget of $\varepsilon_v = 2/255$, step size $\alpha = 0.5/255$, and a total of iterations $S = 10$. The scaling factors are set to $\Lambda = \{0.50,\ 0.75,\ 1.00,\ 1.25,\ 1.50\}$. For baseline methods, method-specific hyperparameters follow their original papers when applicable, while budgets are aligned across methods.

**Metrics**. Attack effectiveness is measured by the Attack Success Rate (ASR), defined as the drop in Rank-1 retrieval accuracy (R@1) over queries correctly retrieved before attack. An attack is successful if the correct match is no longer ranked first after perturbation. Results are reported for both text-to-image (TR R@1) and image-to-text (IR R@1) retrieval.

### B. Main Experiments

Table II reports the attack success rates (ASR) on Flickr30K and MSCOCO. We first evaluate transferability within shared architectural paradigms, including fusion-based models (ALBEF and TCL) and alignment-based models ($\text{CLIP}_{\text{ViT}}$ and $\text{CLIP}_{\text{CNN}}$). On Flickr30K, under ALBEF $\rightarrow$ TCL, SRGA improves over SGA by 3.88% in TR R@1 and 2.19% in IR R@1, while remaining competitive with the strongest baselines. Under TCL $\rightarrow$ ALBEF, SRGA surpasses DRA by 2.97% in TR R@1 and 5.00% in IR R@1, while remaining competitive

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TABLE II: Comparison of attack transferability on Flickr30K (left) and MSCOCO (right). ASR (%) is reported as the drop in R@1. Gray-shaded entries denote white-box attacks, and the remaining entries are black-box transfer results.

| Source | Attack | Flickr30K | | | | | | | | MSCOCO | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ALBEF | | TCL | | $\text{CLIP}_{\text{ViT}}$ | | $\text{CLIP}_{\text{CNN}}$ | | ALBEF | | TCL | | $\text{CLIP}_{\text{ViT}}$ | | $\text{CLIP}_{\text{CNN}}$ | |
| | | TR | IR | TR | IR | TR | IR | TR | IR | TR | IR | TR | IR | TR | IR | TR | IR |
| **ALBEF** | PGD | 52.45* | 58.65* | 3.06 | 6.79 | 8.96 | 13.21 | 10.34 | 14.65 | 76.70* | 86.30* | 12.46 | 17.77 | 13.96 | 23.10 | 17.45 | 23.54 |
| | BERT-Attack | 11.57* | 27.46* | 12.64 | 28.07 | 29.33 | 43.17 | 32.69 | 46.11 | 24.39* | 36.13* | 24.34 | 33.39 | 44.94 | 52.28 | 47.73 | 54.75 |
| | Sep-Attack | 65.69* | 73.95* | 17.60 | 32.95 | 31.17 | 45.23 | 32.82 | 45.89 | 82.60* | 89.88* | 32.83 | 42.92 | 44.03 | 54.46 | 46.96 | 55.88 |
| | Co-Attack | 77.19* | 83.86* | 15.21 | 29.49 | 23.60 | 36.48 | 25.12 | 38.89 | 79.87* | 87.83* | 32.62 | 43.09 | 44.89 | 54.75 | 47.30 | 55.64 |
| | SGA | 97.50* | 97.19* | 44.89 | 55.60 | 33.74 | 44.72 | 34.74 | 46.15 | 96.75* | 96.95* | 58.56 | 65.38 | 57.08 | 65.25 | 58.95 | 66.52 |
| | DRA | 96.45* | 96.49* | 48.58 | 57.57 | 35.83 | 47.74 | 38.83 | 49.95 | 96.39* | 96.51* | 60.03 | 66.65 | 59.48 | 67.18 | 61.14 | 68.45 |
| | SA-AET | 96.98* | 97.43* | **55.22** | **64.40** | 38.04 | 48.51 | 40.61 | 50.57 | **98.27*** | **98.52*** | **68.19** | **73.99** | 61.50 | 67.31 | 62.29 | 68.78 |
| | SRGA | **97.71*** | **97.84*** | 48.97 | 57.90 | **38.55** | **48.68** | **41.23** | **51.70** | 97.52* | 97.61* | 64.37 | 69.59 | **62.66** | **67.99** | **63.69** | **70.17** |
| **TCL** | PGD | 6.15 | 10.78 | 77.87* | 79.48* | 7.48 | 13.72 | 10.34 | 15.33 | 10.83 | 16.52 | 59.58* | 69.53* | 14.23 | 22.28 | 17.25 | 23.12 |
| | BERT-Attack | 11.89 | 26.82 | 14.54* | 29.17* | 29.69 | 44.49 | 33.46 | 46.07 | 35.32 | 45.92 | 38.54* | 48.48* | 51.09 | 58.80 | 52.23 | 61.26 |
| | Sep-Attack | 20.13 | 36.48 | 84.72* | 86.07* | 31.29 | 44.65 | 33.33 | 45.80 | 41.71 | 52.97 | 70.32* | 78.97* | 50.74 | 60.13 | 51.90 | 61.26 |
| | Co-Attack | 23.15 | 40.04 | 77.94* | 85.59* | 27.85 | 41.19 | 30.74 | 44.11 | 46.08 | 57.09 | 85.38* | 91.39* | 51.62 | 60.46 | 52.13 | 62.49 |
| | SGA | 48.91 | 59.68 | 98.52* | 98.79* | 34.00 | 45.20 | 38.57 | 47.92 | 65.93 | 73.30 | 98.97* | 99.15* | 56.34 | 63.99 | 59.44 | 65.70 |
| | DRA | 51.20 | 61.62 | 98.10* | 98.38* | 36.56 | 48.20 | 41.25 | 50.81 | 67.80 | 75.59 | 98.84* | 99.01* | 60.85 | 67.27 | 63.02 | 69.28 |
| | SA-AET | **55.06** | 65.83 | 98.10* | 98.07* | 36.69 | 47.19 | 42.78 | 50.60 | **71.49** | **78.40** | 98.20* | 98.46* | **62.19** | 68.58 | 64.00 | 70.47 |
| | SRGA | 54.33 | **66.80** | **99.37*** | **99.31*** | **38.16** | **49.12** | **44.73** | **51.84** | 71.32 | 78.36 | **99.37*** | **99.49*** | 61.42 | **68.78** | **64.96** | **71.11** |
| **$\text{CLIP}_{\text{ViT}}$** | PGD | 2.50 | 4.93 | 4.85 | 8.17 | 70.92* | 78.61* | 5.36 | 8.44 | 7.24 | 10.75 | 10.19 | 13.74 | 54.79* | 66.85* | 7.32 | 11.34 |
| | BERT-Attack | 9.59 | 22.64 | 11.80 | 25.07 | 28.34* | 39.08* | 30.40 | 37.43 | 20.34 | 29.74 | 21.08 | 29.61 | 45.06* | 51.68* | 44.54 | 53.72 |
| | Sep-Attack | 9.59 | 23.25 | 11.38 | 25.06 | 79.75* | 86.79* | 30.78 | 39.76 | 23.41 | 34.61 | 25.77 | 36.84 | 68.52* | 77.94* | 43.11 | 49.76 |
| | Co-Attack | 10.57 | 24.33 | 11.94 | 26.69 | 93.25* | 95.86* | 32.52 | 41.82 | 30.28 | 42.67 | 32.84 | 44.69 | 97.98* | 98.80* | 55.08 | 62.51 |
| | SGA | 12.83 | 27.10 | 13.91 | 29.10 | 99.02* | 99.03* | 39.59 | 47.99 | 33.41 | 44.64 | 37.54 | 47.76 | 99.79* | 99.79* | 58.93 | 65.83 |
| | DRA | 12.72 | 29.44 | 14.12 | 31.05 | 99.02* | 98.97* | 44.57 | 51.18 | 35.71 | 47.84 | 36.88 | 48.52 | 99.66* | 99.72* | 64.16 | 70.39 |
| | SA-AET | 14.18 | 30.22 | 15.49 | 30.88 | 98.77* | 99.16* | 46.36 | 50.50 | 36.58 | 48.91 | 36.94 | 49.49 | 99.31* | 99.66* | 64.81 | 70.79 |
| | SRGA | **14.39** | **30.87** | **16.76** | **31.79** | **99.39*** | **99.32*** | **49.84** | **54.65** | **39.93** | **49.91** | **39.34** | **51.26** | **99.81*** | **99.87*** | **70.17** | **74.30** |
| **$\text{CLIP}_{\text{CNN}}$** | PGD | 2.09 | 4.82 | 4.00 | 7.81 | 1.10 | 6.60 | 86.46* | 92.25* | 7.01 | 10.62 | 10.08 | 13.65 | 4.88 | 10.70 | 76.99* | 84.20* |
| | BERT-Attack | 8.86 | 23.27 | 12.33 | 25.48 | 27.12 | 37.44 | 30.40* | 40.10* | 23.38 | 34.64 | 24.58 | 29.61 | 51.28 | 57.49 | 54.43* | 62.17* |
| | Sep-Attack | 8.55 | 23.41 | 12.64 | 26.12 | 28.34 | 39.43 | 91.44* | 95.44* | 26.53 | 39.29 | 30.26 | 41.51 | 50.44 | 57.11 | 88.72* | 92.49* |
| | Co-Attack | 8.79 | 23.73 | 13.10 | 26.07 | 28.79 | 40.03 | 94.76* | 96.89* | 29.83 | 41.97 | 32.97 | 43.72 | 53.10 | 58.90 | 96.72* | 98.56* |
| | SGA | 10.32 | 24.18 | 13.70 | 24.89 | 32.75 | 42.14 | 99.49* | 99.45* | 31.61 | 43.00 | 34.81 | 45.95 | 56.62 | 60.77 | 99.61* | 99.80* |
| | DRA | **12.41** | 26.71 | 14.54 | 29.24 | 33.50 | 46.17 | 98.85* | 99.11* | 33.02 | 44.83 | 33.60 | 46.22 | 59.40 | 64.44 | 99.47* | 99.71* |
| | SA-AET | **12.41** | 27.36 | 14.44 | 29.48 | 35.09 | 45.72 | **99.87*** | 99.52* | 33.51 | 45.31 | 34.81 | 46.76 | 60.51 | 64.44 | 99.51* | 99.67* |
| | SRGA | 12.16 | **27.74** | **14.90** | **30.31** | **41.96** | **49.77** | 99.62* | **99.52*** | **35.29** | **47.54** | **37.68** | **49.17** | **65.01** | **69.15** | **99.69*** | **99.89*** |

TABLE III: Ablation results (%) of SRGA.

| Method | ALBEF | | TCL | | $\text{CLIP}_{\text{ViT}}$ | | $\text{CLIP}_{\text{CNN}}$ | |
|---|---|---|---|---|---|---|---|---|
| | TR R@1 | IR R@1 | TR R@1 | IR R@1 | TR R@1 | IR R@1 | TR R@1 | IR R@1 |
| w/o LT | 12.30 | 29.14 | 15.07 | 29.69 | 98.71* | 98.64* | 45.85 | 51.26 |
| w/o SND | 11.51 | 29.21 | 14.96 | 29.62 | 97.53* | 97.63* | 45.62 | 50.46 |
| w/o WM | 12.41 | 29.28 | 15.01 | 29.62 | 98.65* | 98.24* | 46.23 | 51.60 |
| w/o Image | 9.49 | 23.13 | 10.43 | 24.79 | 30.55* | 39.05* | 27.84 | 36.05 |
| w/o Text | 4.07 | 7.76 | 5.37 | 10.45 | 96.93* | 98.00* | 14.18 | 16.30 |
| Completed | **14.32** | **30.60** | **16.46** | **31.54** | **99.58*** | **99.50*** | **49.65** | **54.07** |

TABLE IV: Runtime Comparison (hh:mm:ss).

| Method | ALBEF | TCL | $\text{CLIP}_{\text{ViT}}$ | $\text{CLIP}_{\text{CNN}}$ |
|---|---|---|---|---|
| SGA | 0:28:11 | 0:27:30 | 0:39:53 | 0:40:38 |
| DRA | 0:46:34 | 0:46:58 | 0:59:15 | 0:58:54 |
| SA-AET | 1:19:31 | 1:19:24 | 1:28:57 | 1:26:52 |
| SRGA (Ours) | **0:42:30** | **0:41:20** | **0:52:06** | **0:51:03** |

with SA-AET. For alignment-based transfer between $\text{CLIP}_{\text{ViT}}$ and $\text{CLIP}_{\text{CNN}}$, SRGA outperforms SA-AET by 3.29%–6.40% in ASR. This suggests that detector-derived semantic guidance is more effective when surrogate-specific saliency cues are less reliable across heterogeneous encoders.

Second, we evaluate transferability across architectural paradigms between fusion- and alignment-based models. When using ALBEF as the source and alignment-based models ($\text{CLIP}_{\text{ViT}}$ and $\text{CLIP}_{\text{CNN}}$) as targets, SRGA remains competitive with SA-AET and achieves slight ASR gains of 0.09% to 0.96% in some settings. In the reverse setting ($\text{CLIP}_{\text{ViT}} \rightarrow$ ALBEF), SRGA also shows marginal improvements over SA-AET, with TR and IR increasing by 0.14% and 0.38%, respectively. Similar advantages are also observed on MSCOCO. These results show that detector-derived semantic regions can serve as an effective guidance source for transferable attacks across different VLP architectures.

### C. Discussion

*1) Ablation Study.* SRGA includes three main components, namely Local Transformation (LT), Semantic Normalization Dictionary (SND), and Weighting Mask (WM). We further analyze the roles of text and image perturbations through the *w/o Text* and *w/o Image* settings. Table III shows that removing any component or either perturbation modality reduces transferability, while the full SRGA performs best. Although the text perturbation budget is limited to $\epsilon_t = 1$, a single token substitution may still introduce some semantic shift in the image-caption pair, which is a limitation of this setting.

*2) Computational Efficiency.* We compare the runtime of SGA, DRA, SA-AET, and SRGA under the same iteration budget. The runtime results are reported on Flickr30K with a batch size of 2, and the runtime of SRGA includes detector-based region selection. As shown in Table IV, SRGA is substantially more efficient than SA-AET, reducing runtime by 43%–55%. These results indicate that the proposed guidance mechanism introduces only modest computational overhead in practice.

## IV. Conclusion

In this letter, we propose SRGA, a transferable attack framework that replaces surrogate-specific saliency cues with detector-derived semantic regions for cross-model guidance in VLP models. By using these regions to guide coordinated perturbations in both image and text modalities, SRGA achieves competitive black-box transferability across diverse architectures with favorable efficiency, although its guidance is limited by detector category coverage and may be less effective in open-vocabulary settings. Future work will explore more accurate phrase-region grounding to further strengthen semantic guidance in transferable attacks.

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